过程失效模式及后果分析(PFMEA)更改履历表

客户名称：联电　　客户零件名称：后端盖

变更前版本：00　　FORYOU P/N：XXX-XXXX

变更后版本：01　　客　户 P/N：F01RXXXXXX

变更日期	变更内容	变更原由	备注
2023/1/10	第一版	初版发行	

其它部门会签：

质量部：***　　制造1,2,3部：*** ，*** ，*** ，***

制　作：　　日期：2023/1/10

确　认：　　日期：

承　认：　　日期：2023/1/10

CC：制造1，2，3部、质量部、资料室、项目部

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS (潜在失效模式及影响分析)

Part Description(产品名称) : 后端盖

Part No产品编号:F01RXXXXXX-XXXX

FMEA's Issue No.编号: FM-XXX-1

Model /Vehicle 车型/车辆 :

Key Date关键日期: 2022-12-23

Date修改日期: 2023/1/10 Rev版本: 01

Designer设计責任人:

D/PFMEA Date设计/过程制作日期: 2022-8-19

Page 业数: 11

Core Functional Team多功能小组: *** *** *** *** *** *** *** *** *** *** *** *** *** *** ***

Prepared By 编制者: ***

CC副本:

ID	Process Step / Function 过程步骤/功能	Requirement 要求	Potential Failure Mode 潜在失效模式	Potential Effect(s) of Failure 潜在失效影响	Severity 严重度	Classification 分类	Potential Cause(s) of Failure 潜在失效原因	Occurrence 频度	Current Process Controls Prevention 现行过程控制 预防	Current Process Controls Detection 现行过程控制 探测	Detection 探测度	RPN 风险顺序数	Recommended Action 推荐措施	Responsibility & Target Completion Date 职责和目标完成日期	Action Results实施结果				
															Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
Q1	Incoming Material Inspection 进料检验	Alloy AlSi12Fe can meet the composition AlSi12Fe化学成分符合要求	Alloy composition Deviation from AlSi12Fe 铝料的成份不符合AlSi12Fe 成分要求	Ingot chemical composition out of specification as part physical integrity affected 铝料成份不合格影响产品的整体性能	8		Incorrect alloy from supplier / Composition out of tolerance/Wrong color code 進錯料, 材料成分超差或顏色標識錯誤	2	Marking with appropriate color to incoming ingote 對進料鋁錠加以適當顏色標識	Ingot composition analysis per lot / For all unqualified material must be returned to supplier, refer to standard WI for Material content Inspection 每个炉号进行成份分析，所有不合格的材料一律退回供应商处。参照材料成份检查标准操作指引。	5	80	None 无							
		Aluminum should be free from mouldy 铝料不能发霉	Aluminum ingot was mouldy 铝料发霉.	When melting AL material, Hydrogen in the molten AL which cause porosity in casting process，it will affect the usage peformance.. 铝料熔炼时，铝液中含有氢气，使得压铸時產生砂孔影响产品的使用性能。	8		Bad environment cause mouldy 恶劣的环境使铝料发霉	2	1.Improve the store envirement, and keep it dry. 改善铝料存放的环境，且保持干燥； 2. To ensure during the transportation the Aluminum will be kept dry. 确保运输过程干燥	follow the WI to do the inspection 按照检查基准书进行检查。	5	80	None 无							
		Aluminum should be free from impurities 铝料不能有杂质	Aluminum have impurities. 铝料有杂质；	affect the product peformance 1. 影响产品性能 The cutter will be broken and then cause the dim. Out of spec.. 2. 导致加工刀具崩缺造成产品尺寸不良	8		Bad store and transport environment was not clean. 恶劣的存储和运输环境存在杂质	2	Improve the store and transport envirement to keep it clean; 改善铝料存放的环境，且保持干净； 1.To control the cleanliness of store and transport place.. 控制存储地和运输过程中保持干净。	follow the WI to do the inspection 按照检查基准书进行检查。	5	80	None 无							
		The package cleanliness is OK. 包装是好的	The packing is smudged and crumbled 包装有脏污、碎屑	Affect the cleanliness of the part. 包装使产品的清洁度差	6		The packing film of the incoming material from the supplier is damaged. 供应商来料包装的围膜破损，	2	Train the staff to check the cleanliness of packaging during incoming inspection 进料检验员工检查包装清洁度	1. Sampling inspection of each batch of incoming materials 2. 100% confirmation will be made when the packaging material is used online 1.每批来料抽检 2.包材上线使用的时候100%确认	5	60	None 无							

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		Φ8 pin size qualified Φ8销钉尺寸合格	dimension deviation 尺寸超差,	Affect the assembly 影响产品组装	6		sub supplier manufacture problem cause dimension NOK, mix products 供应商生产过程加工不良，产品混料	2	Sampling inspection by Foryou Foryou来料检查	follow the WI to do the inspection 按照检查基准书进行检查。	4	48	None 無					
		Φ5 pin size qualified Φ5销钉尺寸合格	dimension deviation 尺寸超差,	Affect the assembly 影响产品组装	6		sub supplier manufacture problem cause dimension NOK, mix products 供应商生产过程加工不良，产品混料	2	Sampling inspection by Foryou Foryou来料检查	follow the WI to do the inspection 按照检查基准书进行检查。	4	48	None 無					
		Bush size qualified 衬套尺寸合格	dimension deviation 尺寸超差,	Affect the assembly 影响产品组装	6		sub supplier manufacture problem cause dimension NOK, mix products 供应商生产过程加工不良，产品混料	2	Sampling inspection by Foryou Foryou来料检查	follow the WI to do the inspection 按照检查基准书进行检查。	4	48	None 無					
		Base coating dosage form number meets requirements底涂剂型号符合要求	The base coating dosage form does not meet the requirements底涂剂型号不符合要求	Affect product performance 影响产品性能	6		Supplier incoming material error供应商来料错误	2	Sampling inspection by Foryou Foryou来料检查	follow the WI to do the inspection 按照检查基准书进行检查。	3	36	None 無					
10	Storage 存储	Alloy AlSi12Fe can meet the composition AlSi12Fe化学成分符合要求	Alloy composition Deviation from AlSi12Fe 鋁料的成份不符合AlSi12Fe 成分要求	Ingot chemical composition out of specification as part physical integrity affected 铝料成份不合格影响产品的整体性能	8		Missing or misplaced color identification cause the mix material. 颜色标识漏放、错放导致材料混料	2	Detail label, color mark, storage location are used for different ingot. 不同的铝锭有不同的颜色标签，颜色标记合理存放	1. 每次取原材料要确认铝锭上的颜色和回收料桶的颜色 2.对每炉材料进行成分检测	5	80	None 无					
		Aluminum should be free from impurities 铝料不能有杂质	Aluminum have impurities. 铝料有杂质；	affect the product peformance 1. 影响产品性能 The cutter will be broken and then cause the dim. Out of spec.. 2. 导致加工刀具崩缺造成产品尺寸不良	8		Bad store and transport environment was not clean. 恶劣的存储和运输环境存在杂质	2	Improve the store and transport envirement to keep it clean; 改善铝料存放的环境，且保持干燥与干净；	1.To control the cleanliness of store and transport place.. 控制存储地和运输过程中保持干净。 2.follow the WI to do the inspection 按照检查基准书进行检查。	5	80	None 无					
		The pin must not be contaminated 销钉不能有污染	The pin is contaminated销钉有污染	affect the product cleanliness 1. 影响产品清洁度	8		Bad store and transport environment was not clean. 恶劣的存储和运输环境	2	Improve storage environment, and keep dry and clean; 改善存放的环境，且保持干燥与干净；	1.To control the cleanliness of store and transport place.. 控制存储地和运输过程中保持干净。 2.follow the WI to do the inspection 按照检查基准书进行检查。	5	80	None 无					
		The bushing should not be contaminated 衬套不能有污染	The bushing is contaminated衬套有污染	affect the product cleanliness 1. 影响产品清洁度	8		Bad store and transport environment was not clean. 恶劣的存储和运输环境	2	Improve storage environment, and keep dry and clean; 改善存放的环境，且保持干燥与干净；	1.To control the cleanliness of store and transport place.. 控制存储地和运输过程中保持干净。 2.follow the WI to do the inspection 按照检查基准书进行检查。	5	80	None 无					
		Store the base coat sealed 底涂剂密封存储	Primer is not sealed底涂剂没密封	affect the product peformance 1. 影响产品性能	8		瓶子打开后没有拧紧密封	2	Improve storage environment, and keep dry and clean; 改善存放的环境，且保持干燥与干净；	1.To control the cleanliness of store and transport place.. 控制存储地和运输过程中保持干净。 2.follow the WI to do the inspection 按照检查基准书进行检查。	5	80	None 无					

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20	Material Melting 熔料	Alloy AlSi12Fe can meet the composition AlSi12Fe化学成分符合要求	Chemical composition of Melted AL are out of Spec. 鋁液的化學成份不合格	Part chemical composition out of specification, parts scraped 100%. 產品化學成份不合格，導致產品100%報廢	8		1. Incorrect recycle material 1.未正确回收铝料	2	1. training the operator use the correct recycle material containers with the right color. 1.培训员工正确使用有颜色标记回收筒 The cake is engraved with the brand of the material 2.料饼上刻上材料的牌号	Chemical composition is checked in every furnace. 每炉进行成份测试。	5	80	None 無						
		Alloy AlSi12Fe can meet the composition AlSi12Fe化学成分符合要求	Chemical composition of Melted AL are out of Spec. 鋁液的化學成份不合格	Part chemical composition out of specification, parts scraped 100%. 產品化學成份不合格，導致產品100%報廢	8		2. Material Mixed in the recycle container 2.回收筒中的回收料混料	2	1. training the operator use the correct recycle material containers with the right color. 1.培训员工正确使用有颜色标记回收筒 The cake is engraved with the brand of the material 2.料饼上刻上材料的牌号	Chemical composition is checked in every furnace. 每炉进行成份测试。	5	80	None 無						
		Alloy AlSi12Fe can meet the composition AlSi12Fe化学成分符合要求	Chemical composition of Melted AL are out of Spec. 鋁液的化學成份不合格	Part chemical composition out of specification, parts scraped 100%. 產品化學成份不合格，導致產品100%報廢	8		4. Melting wrong ingot 4.熔错料	2	4.Alloy is made the color mark 4.铝锭用颜色标识	Chemical composition is checked in every furnace 每炉进行成份测试	5	80	None 無						
		Aluminum should be free from impurities 铝料不能有杂质	Aluminum have impurities. 铝料有杂质；	affect the product peformance 1. 影响产品性能 The cutter will be broken and then cause the dim. Out of spec.. 2. 导致加工刀具崩缺造成产品尺寸不良	8		3. Foreign debris in recycle material 3.回收料中有其它杂质	2	Improve the store and transport envirement to keep it clean; 改善铝料存放的环境，且保持干净； Refining and slagging shall be carried out according to the work instruction 1.按照作业指导书进行精炼打渣	Use K mould to check the impurities in every furnace. 每炉做K模测试。	5	80	None 無						
			Excessive dregs in molten material 鋁液中有多余的渣滓	When melting AL material,excessive dregs will cause porosity, it will affect the usage peformance. 铝料熔炼时，铝液中含有铝渣压铸时產生砂孔影响产品的使用性能。	8		1.Utilize inadequate amount de-dregs compound. 1.使用的打碴剂不夠量 2. Adding aluminum after no Skimming 2. 加入铝料后未扒渣	2	1.Accord alloy weight balance de-dregs compound and record 每次對打碴剂稱重並記錄 2. Per adding aluminum after close furnace door 3~5 minutes then skimming 2.每加一次料后关闭熔炉3~5分钟后扒渣	1.Check Alloy sample every lot 取樣看鋁料斷口 2. K-mold inspect 2.K模检查	5	80	None 無						
					8		3.Molten AL temperature out of Spec. 3.熔料温度不在规格内.	2	Monitoring the temperature continuously 對溫度進行持續控制。	Record each melting process. 每爐熔料記錄	5	80	None 無						
		No excessive gas in molten material 鋁液中没有气体	Excessive gas in molten material 鋁液中有多余的气体	The gas will cause porosity during the casting, it will affect the usage peformance. 鋁液中含有气体会造成铸件中有气孔，气孔会影响产品的使用性能	7		Improper de-gassed time flow pressure , flux. 除气机時間及流量不對和气压不足	2	1.Degassing each transfer package, Set up the flow and pressure of the Argon used in degasing. 每转运包除气，設置除气机的气压和時間，流量 2.Hydrogen inspect 2.含氢量检测	Check de-gassing data each time 檢查每次除气的数据	5	70	None 無						
30	Die casting 压铸	Aluminum temperature in the spec. (650~680℃) 铝液的温度在规格范围内	Molten AL temperature in holding furnace is out of Spec. (650~680℃) 爐中鋁液的溫度未在650~680℃之間	The temperature was not in the spec. the product will affect the usage peformance. 鋁液溫度达不到要求会影响产品的使用性能	7		1. Failure of furnace temperature monitor facility. 熔爐的溫度控制器出故障。	3	1. Regular calibration of furnace temperature monitor facility. 1.對熔爐溫度控制器進行日常校准。 Monitoring the temperature continuously 對溫度進行持續控制。	every four hours check by the operator 每4小时确认料温 set up the alarm when the temperature was over 680℃ or blow 650℃ by the machine. 温度超过680℃或低于650℃设备自动报警。	4	84	None 無						
									2. Regular equipment maintained. 2.對設備進行日常保養。		4	84	None 無						
								3	3.Regular maintenance of thermocouple 3.对测量温度的热电偶定期维护校准	Outgoing inspection and calibration are issued annually 每年外发点检	4	84	None 無						
							2. Burner does not work well. 2. 火爐工作異常。	3	1. Regular calibration of furnace temperature monitor. 1.對熔爐溫度控制器進行日常校准。	set up the alarm when the temperature was over 680℃ or blow 650℃ by the machine. 温度超过680℃或低于650℃设备自动报警。	4	84	None 無						
									2. Regular equipment maintenance . 2.對設備進行日常保養。		4	84	None 無						

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		appearance meet the specification. 外觀符合要求.	appearance can't meet the specification. 外觀不符合要求.	Non-confirming (cosmetic)part, cause to Internal scrap part 產品的外觀不合格, 導致產品報廢	7		1. Inadequate die maintenance; 2. The parameter setting is not in accordance with the operation instruction. 3. The setting of spray does not follow the operation instruction 1. 模具沒保養好; 2. 參數设定沒有按照作业指导书。 3. 喷雾的设定沒有按照作业指导书	2	1.1the product engineer and die casting technician check the mould after maitance finish; 1.1压铸技术员与产品工程师检查模具在保养完成后; 1. 2. the die casting operater check whether mould No. is correct and cooling is normal and the surface of mould has crack and others before every setup. 1.2.压铸操作员每次上模前检查模具牌号是否正确，运水是否正常，模具表面是否开裂等; 2.技术员/工程师按照作业指导书设定参数和喷雾	The product techincian check the parameter 1.技术员每6小时点检确认机台参数 2.first product and processing inspection by PQC. PQC对产品作首件和过程检查	5	70	None 無						
		Bushing is leak-free 衬套无漏装	Bushing is missing 衬套漏装	Impact on product function 影响产品功能	7		1, adjust the machine parts mixed with normal products 2. Steel sleeve falls off during die casting 1，调机件混入正常品 2，压铸过程中钢套掉落	2	1. Spray defective product label paint on the parts and dispose of them as scrap 2, after the completion of die casting, each product operator 100% self-check 1，调机件喷上不良品标识油漆，报废处理 2，压铸完成后每件产品作业员100%自检	1.100% visual inspection 1.100%目視检查 2. Patrol check 2.PQC抽查	5	70	None 無						
		The date code on the product is correct/产品上有日期码而且正确	Modl date code is missed or error 产品日期码漏打、错打	Product can't fit the traceability 产品不能追溯或追溯错误	7		1.Not inspection before loading into Casting Machine 1.上模前未检查模具。 2.moudl missing mark date code. 2.模具漏打日期码标记。	3	1.Inspection before loading into Casting Machine. 1. 上模前检查.	first product and patrol inspection by PQC. PQC对产品作首件和过程检查	5	84	None 无						
		the product is filling completly 产品填充良好	the product was not filling incompletly. 产品填充不良	affect the product peformance 影响产品性能	7		1.Wrong casting parameters 1.压铸压力过小。 2.Not enough material 取汤量不足	2	Technician or engnieer follow the WI to set up the parameters before starting die-casting. 技术员在开始压铸前调好合适的压力和取汤量 the first is ok and then start manufacturing. 2.必须首件OK后方可生产。	First piece inspection and patrol inspection by PQC PQC首件检测和巡检	5	70	None 无						
		There is no cold shut on the product. 产品表面无冷隔	There is cold shut on the product. 产品表面有冷隔	affect the product peformance 影响产品性能	7		1.Wrong casting parameters 1.压铸压力过小。 2.Not enough material 取汤量不足 Spray amount is too large, blow not dry 3.喷雾量过大，吹气没有吹干	2	Technician check the parameters before starting die-casting. 技术员在开始压铸前调好合适的压力和取汤量 Follow the WI to set up the parameter of spray. 按照作业指导书书设置喷涂。	First piece inspection and patrol inspection by PQC PQC首件检测和巡检	5	70	None 无						
		Stomata meet the requirements of shipping regulations气孔满足出货规程要求	Sealing surface pores exceed 0.7mm requirements, non-sealed processing surface exceeds 1mm, internal pores 5%/1密封面气孔超出0.7mm要求，非密封加工面超出1mm，内部气孔5%/1	Affect the product performance 影响产品的性能	7		1. Excessive gas content of aluminum material 2. The soup feeder does not feed enough, resulting in thin cake 3. The parameters are not set according to the operation instructions. 1.铝料的含气量过多 2.给汤机给料不足，导致料饼薄 3.没有按照作业指导书设定参数。	2	De gas every transfer ladle 1.每转运包正确的除气 Adjust the amount of soup in the soup feeder 2.调整好给汤机的给汤量 3.Follow the WI to set up the parameter. 按照作业指导书书设置参数。	The product techincian check the parameter 1.技术员每6小时点检确认机台参数 2.Use X-Ray to check the first product and processing inspection by PQC. PQC用X光机对产品作首件和过程检查 3.100% check in the X-Ray 100% check process. 在X光全检工序全检。	5	70	None 无						
		The product was not sticking 产品不粘模	The product was sticking 产品粘模	The product was damage, it will affect the assembly and the usage performance. 产品缺料，影响客户的装配和产品的使用性能	7		1. Insufficient spray quantity leads to excessively high mold temperature in local parts of the mold 2. The mold is under-cut. 3. The ratio and concentration of release agent is too low and the release effect is poor. 1.喷雾量不足导致模具局部的模温过高 2.模具有倒扣 3.脫模剂的配比浓度过低脫模效果差	2	1. Set spray parameters according to the operation instructions 2. Check the mold before loading the mold 3. Automatic proportioning of release agent 1.按照作业指导书设置喷雾参数 2.上模前检查模具 3.脫模剂自动配比	The product techincian check the parameter 1.技术员每6小时点检确认机台参数 2.Use X-Ray to check the first product and processing inspection by PQC. PQC首件检测和巡检	5	70	None 无						

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS (潜在失效模式及影响分析)

Part Description(产品名称) : 后端盖 | Part No产品编号:F01RXXXXXX-XXXX | FMEA's Issue No.编号: FM-XXX-1

Model /Vehicle 车型/车辆 : | Key Date关键日期: 2022-12-23 | Date修改日期: 2023/1/10 | Rev版本: 01

Designer设计责任人: | D/PFMEA Date设计/过程制作日期: 2022-8-19 | Page 业数: 11

Core Functional Team多功能小组: *** *** *** *** *** *** *** *** *** *** *** *** *** *** *** *** | Prepared By 编制者: ***

CC副本:

ID	Process Step / Function 过程步骤/功能	Requirement 要求	Potential Failure Mode 潜在失效模式	Potential Effect(s) of Failure 潜在失效影响	Severity 严重度	Classification 分类	Potential Cause(s) of Failure 潜在失效原因	Occurrence 频度	Current Process Controls Prevention 现行过程控制 预防	Current Process Controls Detection 现行过程控制 探测	Detection 探测度	RPN 风险顺序数	Recommended Action 推荐措施	Responsibility & Target Completion Date 职责和目标完成日期	Action Results实施结果: Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
		The cake of running system thickness is out of spec.(25+/-5mm) 料饼厚度在规格范围内	The cake of running system thickness is less than 13mm 料饼厚度薄过13mm	The product was damage, it will affect the assembly and the usage performance. 产品缺料，影响客户的装配和产品的使用性能	7		1.Wrong setting parameters for soup quantity 1.取汤量设定参数过小	2	1.Set up the correct parameters during the adjustment. 1.调机参数重设取汤量 Cake thickness automatic sorte by machine. 2.料饼厚度自动分选传	The machine 100% check by the machine automatic sorting. 设备自动100%选别 2.Use caliper to check the first product and processing inspection by PQC. PQC首件检测和巡检	6	84	None 无						
		Product location hole for CNC machining didn't have sticking. 产品定位孔无粘模	Product location hole for CNC machining have sticking. 产品定位孔无粘模	Affect the machining location, it will casue the dimensions out of spec. which will affect the assembly. 影响产品的定位造成尺寸不合格，尺寸不合格影响客户的装配。	7		1.The mold structure is aging. 模具结构老化 the local tempereture is too high 2.模具局部温度高 the local spray is not enough. 3.喷雾量不足	2	Instruct the operator to check the mold structure and maintaining after operation 每次下模后检查该结构状态并进行日常维护 the operator check the position and the spray time of the spray. 员工点检确认喷雾位置和喷雾时间。	Technician patrol check the spray parameter every 6 hours 技术员日常点检 the first product and processing inspection by PQC. PQC首件检测和巡检	5	70	None 无						
		Product location hole have not defect 产品定位孔无多料	The locating pore have defect. 产品定位孔粘模、多料。	Affect the machining location, it will casue the dimensions out of spec. which will affect the assembly. 影响产品的定位造成尺寸不合格，尺寸不合格影响客户的装配。	7		The mold structure is aging. 模具结构老化 The Pin of the mold was broken. 模具上的针损坏。	2	1. Instruct the operator to check the mold structure and maintaining after operation 每次下模后检查该结构状态并进行日常维护	the first product and processing inspection by PQC. PQC首件检测和巡检	5	70	None 无						
		Ejector marks deepened less than 0.3mm on the non machining surface. 不加工面顶针印凹入小于0.3mm，	Ejector marks deepened more than 0.3mm on the non machining surface. 不加工面顶针印凹入大于0.3mm	affect the usage performance 影响产品的性能	7		1.The inbalance ejectors cause the deformation of parts. 1.产品压铸时顶出不平衡。 2..产品粘模导致抱紧力大	2	1.The technicican confirms that the mold has not been installed in place. 1.技术员确认模具是否安装到位。 2.Make sure the dimensionof ejecter pin when check the mold. 2.模具检查时确认顶针长度。	the first product and processing inspection by PQC. PQC首件检测和巡检	5	70	None 无						
		No protrusions are allowed on the non machining surface 不加工面顶针印不允许凸出	ejector raised on the non machining surface 不加工面顶针印凸出	Affect the assembly 影响产品组装	7		1.Ejecter wear 2. 顶针磨损。 The push-pull rod is not installed in place 2. 推拉杆安装不到位	2	1.The technicican confirms that the mold has not been installed in place. 1.技术员确认模具是否安装到位。 2.Check the degree of wear and tear 2.每班检查确认顶针是否磨损。	the first product and processing inspection by PQC. PQC首件检测和巡检	3	42	None 无						
		Ejector marks raised on the machining surface 加工面顶针印凸出	Ejector marks sinked into the machining surface 加工面顶针印凹入	ejector mark can't be removed completely, affect the dimension and the usage performance 加工未完全去除，影响尺寸，影响产品的性能	7		1.The inbalance ejectors cause the deformation of parts. 1.产品压铸时顶出不平衡。 The push-pull rod is not installed in place 2. 推拉杆安装不到位	2	1.The technicican confirms that the mold has not been installed in place. 1.技术员确认模具是否安装到位。 2.Check the degree of wear and tear 2.每班检查确认顶针是否磨损。	the first product and processing inspection by PQC. PQC首件检测和巡检	5	70	None 无						
		The flatness 58#，68# of locating surface Max0.5 压铸平面度小于0.5（内控尺寸）	The flatness of locating surface more than 0.5压铸平面度大于0.5（内控尺寸）	it will affect the location of the machine which will cause the dim. Out of tolerance, affect the assembly. 影响产品的定位，引起尺寸不良，影响产品的组装。	7		1. Mould were sticking. 1.模具磨损 2.mbalanced ejection of the thimble 2.顶针顶出不平衡	2	1.polish the mould; 1.抛光模具 2.模具定期维护保养 2.The mould is maintained regularly	PQC patrol inspection according to the process checking instruction PQC按照过程检查指引做巡视检查	5	70	None 无						
40	De-gating 去水口	No breakage along the part edge 边缘没有崩缺	Breakage along the part edge 边缘崩缺	Affects the appearance of the product, the product will be rejected by customer. 影响产品外观，客户拒收	6		Cutter sticked material it will cause breakage when punching 刀具粘料冲切时造成崩缺	3	Operator lubricates according to the operation instruction (once per 20pcs) 操作员按照作业指导书进行打润滑油（once per 20pcs) operator checked the appearance very 20 pcs by himself 员工每20模自检一次	PQC patrol inspection according to the process checking instruction PQC按照过程检查指引做巡视检查	5	90	None 无						
		The remain of burr can meet the specification 0.3Max 毛刺0.3mm以内	The remain of burr can't meet the specification 0.3Max 外观毛刺大于0.3mm	Affect the assembly 影响产品组装	6		the puching cutter wear 冲刀磨损	3	Set up tool life, check and maintain tooling regularly 制定刀具的寿命管理，定期点检和维护保养 operator checked the appearance very 20 pcs by himself 员工每20模自检一次	PQC patrol inspection according to the process checking instruction PQC按照过程检查指引做巡视检查	4	72	None 無						
		The flatness of locating surface no deformation(flatness Max0.2) 产品定位平面无变形	The flatness of locating surface is out of 0.2. 产品定位平面的平面度超差。	Effect the machining dimensions, affect the assembly 影响加工尺寸	7		The support surface didn't blow clean of aluminum 支撑平面铝屑没有吹干净	2	Use the air gun to blow the support surface every punching, operator checked the appearance very 20 pcs by himself 员工每次冲切用气枪吹干净支撑面，每20模自检一次	PQC patrol inspection according to the process checking instruction PQC按照過程檢查指引做巡視檢查	5	70	None 無						
50	De-burring 去毛刺	Appearance can meet the specification. 外觀符合要求.	Appearance cann't meet the specification. 外觀不符合要求.	1.Customer reject the parts 客戶不接受	4		1. The product is polluted and dirty 2. Over-treatment cause damage 1.产品被污染，脏污 2.产品过处理，崩缺	2	training the operator with WI 按作业指导书培训操作员 100% visual inspection by operator. 作业员100%目视检查	PQC patrol inspection according to the process checking instruction PQC按照過程檢查指引做巡視檢查	5	40	None 無						

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS (潜在失效模式及影响分析)

Part Description(产品名称) : 后端盖 | Part No产品编号:F01RXXXXXX-XXXX | FMEA's Issue No.编号: FM-XXX-1

Model /Vehicle 车型/车辆 : | Key Date关键日期: 2022-12-23 | Date修改日期: 2023/1/10 | Rev版本: 01

Designer设计责任人: | D/PFMEA Date设计/过程制作日期: 2022-8-19 | Page 业数: 11

Core Functional Team多功能小组: *** , *** *** *** *** *** *** *** *** *** *** *** *** *** *** , *** | Prepared By 编制者: ***

CC副本:

ID	Process Step / Function 过程步骤/功能	Requirement 要求	Potential Failure Mode 潜在失效模式	Potential Effect(s) of Failure 潜在失效影响	Severity 严重度	Classification 分类	Potential Cause(s) of Failure 潜在失效原因	Occurrence 频度	Current Process Controls Prevention 现行过程控制 预防	Current Process Controls Detection 现行过程控制 探测	Detection 探测度	RPN 风险顺序数	Recommended Action 推荐措施	Responsibility & Target Completion Date 职责和目标完成日期	Action Results实施结果				
															Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
		Appearance can meet the specification. 外观毛刺0.3mm最大	Appearance cann't meet the specification. 外观毛刺高于0.3mm	Affect the assembly 影响产品组装	4		1.Missing operation 漏加工	2	training the operator with WI 按作业指导书培训操作员 100% visual inspection by operator. 作业员100%目视检查	PQC patrol inspection according to the process checking instruction PQC按照过程检查指引做巡视检查	5	40	None 无						
60	Shot blasting 抛丸	The appearance is OK（ the product without skin peeling, missed blasting, dirty，damage 产品外观合格，没有起皮、脏污，漏喷砂，碰伤	The appearance is NOK（ the product with skin peeling, missed blasting, dirty，damage 产品外观不合格，有起皮、脏污，漏喷砂	Out of customer's standard it'll be rejected 不符合客户外观标准，客户拒收	5		1.The sand size is out of spec. 砂粒规格不合要求。 2.Sand blasting machine parameter is wrong. 打砂机参数设置不当。 3. The number of parts for blasting is wrong every time 一次放入产品数量不当。	2	Follow the WI to set up the parameter of shot blasting. 按照作业指导书书设置抛丸。	1. Operator 100% visual inspection 1. 作业员100%目视检查 2. PQC patrol inspection according to the process checking instruction 2. PQC按照过程检查指引做巡视检查	5	50	None 无						
		Roughness Max Rz45 粗糙度符合要求图纸要求RZMax100	Roughness out of spec 粗糙度RZMax100	Out of customer's standard it'll be rejected 不符合客户粗糙度标准，客户拒收	5		1.The steel ball size is out of spec. 砂粒规格不合要求。 2.Sand blasting machine parameter is wrong. 打砂机参数设置不当（时间过长，频率太高，具体参照作业指导书）。	2	Follow the WI to set up the parameter of shot blasting. 按照作业指导书设置抛丸。	1.Operator 100% visual inspection 1. 作业员100%目视检查 2.PQC patrol inspection according to the process checking instruction 2. PQC按照过程检查指引做巡视检查	5	50	None 无						
70	CNC machining and washing beside the machine CNC加工和机边清洗	切削液浓度在5%~10%范围内	切削液浓度低于5%	Poor lubrication during maching, size NG, affect customer assembly 加工时润滑效果不好，尺寸NG，影响客户装配	7		Poor lubrication during maching, resulting in size NG 加工时润滑效果不好，导致尺寸NG	2	Each shift by the special person point check cutting fluid concentration, concentration within the range of specifications can be started 每班由专人点检切削液浓度，浓度在规格范围内方可开机	Check the consistency of machining coolant per 12 hours 每班对 切削液浓度进行点检	5	70	None 无						
		No scratch on the machining surface 加工表面无划痕	Scratch on the machining surface 加工表面划伤	Affect the product performance 影响产品的性能	4		Machining burr is sticked on the tool 刀具粘料	2	Check the consistency of machining coolant each shift 每班检查切削夜浓度	Check the consistency of machining coolant per 12 hours. 每班对切削液浓度进行点检	5	40	None 无						
					4		Insufficient protection of turnover packaging, collision between products 周转包装的防护不足，产品间磕碰	3	Use the special transport package . 使用专用的运输包装方式。	1.PQC patrol check 1.PQC巡检 2.Opertor 100% visual check the function surface. 2.作业员100%检查产品功能面的划伤。	5	60	None 无						
					6		Scratch when taking or placing product 取放产品时划伤	3	Be sure of the protect device when loading fixture 装夹夹具时点检确认防护装置 Trainning the operator 培训员工	1.100% visual inspection by operator. 作业员100%目视检查 2.Spot check by PQC PQC抽查	5	90	None 无						
		No knife mark on the machine surface 加工面无刀纹	knife mark on the machined surface imcompletely 加工面有明显刀纹	Affect the product performance 影响产品的使用性能	5		Tool wear The product is not pressed tight, and the processing is jitter 刀具磨损 产品没有压紧，加工时候抖动	3	1.设定刀具寿命管理，机台自动监控 2.技术员点检夹具压紧	Tool life management, automatic machine monitoring 1.刀具寿命管理，机台自动监控 2.每班技术员点检夹具压紧 3.the first product and processing inspection by PQC. PQC首件检测和巡检	5	75	None 无						
		加工尺寸不超差	加工尺寸超差	affect the assembly and product performance 影响产品组装和性能	7		1.CNC Fixture clamping was not stable. 1.加工夹具没有加紧 2. 夹具的定位销晃动 3.夹具定位面铝屑没有吹干净 4.Cutting tool wear. 4.刀具磨损.	3	1.Inspect the fixuture on time; 1.定期检查夹具; 2.Setup a tool-life record and prepare to have a regular tooling change. 2.定期更换刀具	1.每班技术员点检夹具压紧 2.刀具寿命管理，机台自动监控 3.the first product and processing inspection by PQC. PQC首件检测和巡检	5	105	None 无						
		1. Parts don't have dirty 产品没有脏污	1. Parts have dirty 产品脏污	1.Can't meet cleanliness appearance requirement, affect the assembly and performance 1.达不到清洁度要求,影响客户装配和性能	4		1. Don't clean completely 清洗不彻底 2. Cleaning liquid is dirty 清洗液体脏污	2	1.产品没有完全浸泡到 2.Change the cleaning liqud termly 定期更换清洗液	1. 100% Visual inspection by operator 操作员100% 目视检查 2. Checked by PQC PQC检查	5	40	None 无						

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ID	Process Step / Function 过程步骤/功能	Requirement 要求	Potential Failure Mode 潜在失效模式	Potential Effect(s) of Failure 潜在失效影响	Severity 严重度	Classification 分类	Potential Cause(s) of Failure 潜在失效原因	Occurrence 频度	Current Process Controls Prevention 现行过程控制 预防	Current Process Controls Detection 现行过程控制 探测	Detection 探测度	RPN 风险顺序数	Recommended Action 推荐措施	Responsibility & Target Completion Date 职责和目标完成日期	Action Results实施结果				
															Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
80	Appearance Inspection 外观全检	1. Parts don't have dirty 产品没有脏污	1. Parts have dirty 产品脏污	1.Can't meet cleanliness appearance requirement, affect the assembly and performance 1.达不到清洁度要求,影响客户装配和性能	4		The gloves are dirty and not replaced in time Work without gloves Employees are not familiar with standards 手套脏污未及时更换 未戴手套作业 员工不熟悉标准	3	1.Wear gloves before operating and keep golves clean 1.操作前须戴手套并保证手套清洁 2.Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工 3.Place direct label in proper way 不良品位置放置标识	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	60	None 无						
		No knife mark on the machine surface 加工面无刀纹	knife mark on the machined surface imcompletely 加工面有明显刀纹	Affect the product performance 影响产品的使用性能	5		Operators are not familiar with the standard. 员工不熟悉标准	2	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	50	None 无						
		No scratch on the machining surface 加工表面无划痕	Scratch on the machining surface 加工表面划伤	Affect the product performance 影响产品的性能	6		Operators are not familiar with the standard. 员工不熟悉标准	2	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	60	None 无						
		No burrs on the function surface 产品面无毛刺	Burrs on function surface. 产品面有毛刺。	Affect the product performance 影响产品的性能	6		Operators are not familiar with the standard. 员工不熟悉标准	2	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	60	None 无						
		气孔满足出货规程要求	密封面气孔超出0.7mm要求，非密封加工面超出1mm	Affect the product performance 影响产品的性能	7		Operators are not familiar with the standard. 员工不熟悉标准	2	Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	70	None 无						
		Parts don't have the rust 产品没有发霉	1. Parts have the rust 产品发霉	Affect the product performance 影响产品的性能	6		The product is not thoroughly dried Employees are not familiar with standards 产品未彻底烤干 员工不熟悉标准	3	1.Wear gloves before operating and keep golves clean 1.操作前须戴手套并保证手套清洁 2.Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	90	None 无						
		There is no yellowish mark on the product 产品没有发黄	There is no yellowish mark on the product 产品发黄	Out of customer's standard it'll be rejected 不符合客户外观标准，客户拒收	5		Employees are not familiar with standards 员工不熟悉标准	3	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	75	None 无						
90	Special machine washing & Drying 专机清洗/烘干	满足清洁度图纸要求	清洁度未达到图纸要求	Affect the product usage function，影响客户使用功能	7		1. Washing agent losing concentration. 1.清洗剂浓度不够 2. 超声波发生器故障 3.滤芯没有定期更换 4. 高压喷淋对点偏位	2	1.Chemical concentration checked with record daily. 1.每天对化学浓度进行检测 2.每班对设备进行点检确认 3.规定滤芯更换频率 4.定期点检确认	技术员每班开始生产前点检 Patrol check PQC抽查	5	70	None 无						
		Parts don't have the rust 产品没有发霉	1. Parts have the rust 产品发霉	affect the product performance 影响产品性能	6		1.Don't dry completely 干燥不彻底（时间不够，温度过低）	2	operator follow the WI to dry the products. 作业员按照作业指导书作业	Patrol check PQC抽查	5	60	None 无						
		There is no yellowish mark on the product 产品没有发黄	There is no yellowish mark on the product 产品发黄	Out of customer's standard it'll be rejected 不符合客户外观标准，客户拒收	6		The drying temperature is too high 烘烤的温度过高	2	operator follow the WI to dry the products. 作业员按照作业指导书作业	Patrol check PQC抽查	5	60	None 无						
100	laser marking 激光打标	The DMC can been scaned 二维码清晰可辨别	The DMC can't been scaned 二维码无法辨别	Unable to scan code assembly, affecting assembly 无法扫码装配，影响装配	7		1. The laser marking machine engraving distance is too far, engraving is not clear 2. The surface of the product is dirty 3. Dust in the laser lens 1. 激光打标机刻印的距离太远，刻印不够清晰 2. 产品表面脏污 3. 激光发射器镜头有灰尘	2	1,100% check the scannor whether it can work. 1.扫描仪100%扫描确认是否可以读取。	1.100% inspection by scanner 1.100%扫描检查 2.The assembly process works after 100% scanning code, NG product alarm 2.组装工序100%扫码后作业，NG产品报警	3	42	None 无						

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS (潜在失效模式及影响分析)

Part Description(产品名称) : 后端盖

Part No产品编号:F01RXXXXXX-XXXX

FMEA's Issue No.编号: FM-XXX-1

Model /Vehicle 车型/车辆 :

Key Date关键日期: 2022-12-23

Date修改日期: 2023/1/10 Rev版本: 01

Designer设计责任人:

D/PFMEA Date设计/过程制作日期: 2022-8-19

Page 业数: 11

Core Functional Team多功能小组: *** *** *** *** *** *** *** *** *** *** *** *** *** *** *** ***

Prepared By 编制者: ***

CC副本:

ID	Process Step / Function 过程步骤/功能	Requirement 要求	Potential Failure Mode 潜在失效模式	Potential Effect(s) of Failure 潜在失效影响	Severity 严重度	Classification 分类	Potential Cause(s) of Failure 潜在失效原因	Occurrence 频度	Current Process Controls Prevention 现行过程控制 预防	Current Process Controls Detection 现行过程控制 探测	Detection 探测度	RPN 风险顺序数	Recommended Action 推荐措施	Responsibility & Target Completion Date 职责和目标完成日期	Action Results实施结果				
															Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
		The DMC code is grade B 二维码等级B级	The DMC code is lower than grade B 二维码等级低于B级	Unable to scan code assembly, affecting assembly 无法扫码装配，影响装配	7		1. The laser marking machine engraving distance is too far, engraving is not clear 2. The surface of the product is dirty 3. Dust in the laser lens 1.激光打标机刻印的距离太远，刻印不够清晰 2.产品表面脏污 3.激光发射器镜头有灰尘	2	1,scan by the camera when the machine start 100%. 1.扫描仪100%扫描确认是否可以读取。 2.The printed DMC shall meet the requirements of class B 2.刻印的DMC要达到B级的要求	1.100% inspection by scanner 1.100%扫描检查 2.The assembly process works after 100% scanning code, NG product alarm 2.组装工序100%扫码后作业，NG产品报警	3	42	None 无						
		The information of the QR code is stored in the database 二维码的信息存入数据库	The information of the QR code is not stored in the database 二维码的信息没有存入数据库	Unable to scan code assembly, affecting assembly 无法扫码装配，影响装配	7		1. The network is not good 2. Equipment failure 1.网络不畅通 2.设备故障	2	1. The device saves data on the local hard disk 2. The software automatically recognize and input the data into the database 3. Software alarm 1,设备保存数据在本地硬盘 2.软件自动抓取数据录入数据库 3.软件报警	1.100% inspection by scanner 1.100%扫描检查 2.The assembly process works after 100% scanning code, NG product alarm 2.组装工序100%扫码后作业，NG产品报警	3	42	None 无						
		the two-dimensional code must be engraving 100% on products 产品要刻100%刻二维码	no two-dimensional code on one product 产品没有二维码	Unable to scan code assembly, affecting assembly 无法扫码装配，影响装配	7		Missing operation 漏作业	2	1,scan by the camera when the machine start 100%. 扫描仪100%扫描确认是否可以读取。	1.100% inspection by scanner 1.100%扫描检查 2.The assembly process works after 100% scanning code, NG product alarm 2.组装工序100%扫码后作业，NG产品报警	3	42	None 无						
		code is not duplicated 二维码没有重复刻码(两个码打在一个产品上)	code is duplicated 二维码重码(两个码打在一个产品上)	Unable to scan code assembly, affecting assembly 无法扫码装配，影响装配	7		operation repeats 重复作业	2	1,scan by the camera when the machine start 100%. 扫描仪100%扫描确认是否可以读取。	1.100% inspection by scanner 1.100%扫描检查 2.The assembly process works after 100% scanning code, NG product alarm 2.组装工序100%扫码后作业，NG产品报警	3	42	None 无						
		the code is not corresponding to the part 二维码没有重码	Two or more products share the same code 两个或多个产品使用同一个二维码	Unable to scan code assembly, affecting assembly 无法扫码装配，影响装配	7		Equipment failure, no code updated 设备故障，刻码没有跳码	2	1,scan by the camera when the machine start 100%. 1.扫描仪100%扫描确认是否可以读取。 2.Compare the DMC data when entering the database. If the duplicate code cannot be recorded, alarm will be given 2.DMC数据录入数据库时做对比，重复码的不可录入并报警	1.100% inspection by scanner 1.100%扫描检查 2.The assembly process works after 100% scanning code, NG product alarm 2.组装工序100%扫码后作业，NG产品报警	3	42	None 无						
110	Leakage test 测漏	meet the drawing leakage testrequirement 满足图纸的测漏要求	Cavity one leak overa 0.8 Norm-ml/min 一腔测漏超过0.8 Norm-ml/min	Part scrap in supplier 零件在内部报废	7		1. sealing area was sticking,the sealing surface can't seal. 密封位置粘料导致密封不住 2. sealing area was demaged. 密封圈位置被破坏 3. the part was deformation 产品变形 4.sealing gasket was broken 密封圈损坏 5.the leakage test equipment test system failure 测漏设备故障	2	1. Piont impact by leak test machine automatic when part pass test OK品自动打点 2. Use NG sample for spot check 使用NG样件进行点检 Make MSA during making sample. 制作样品时制作MSA.	1.Operator 100% visual inspection 1.作业员100%目视检查 2.PQC patrol inspection according to the process checking instruction 2.PQC按照过程检查指引做巡视检查	4	56	None 无						
		meet the drawing leakage testrequirement 满足图纸的测漏要求	Cavity one leak overa 0.8 Norm-ml/min 二腔测漏超过0.8 Norm-ml/min	Part scrap in supplier 零件在内部报废	7		1. sealing area was sticking,the sealing surface can't seal. 密封位置粘料导致密封不住 2. sealing area was demaged. 密封圈位置被破坏 3. the part was deformation 产品变形 4.sealing gasket was broken 密封圈损坏 5.the leakage test equipment test system failure 测漏设备故障	2	1. Piont impact by leak test machine automatic when part pass test OK品自动打点 2. Use NG sample for spot check 使用NG样件进行点检 Make MSA during making sample. 制作样品时制作MSA.	1.Operator 100% visual inspection 1.作业员100%目视检查 2.PQC patrol inspection according to the process checking instruction 2.PQC按照过程检查指引做巡视检查	4	56	None 无						
		The leakage pressure meets the test requirements 测漏压力符合检测要求	Leakage pressure does not meet test requirements 测漏压力不符合检测要求	Effect customer usage. 影响客户使用功能	7		the leakage test equipment test system failure 测漏设备故障	3	1. Calibrate leak detection equipment with OK, NG samples for each shift. 2. The equipment automatically monitors and measures the air pressure, and alarms when it fails to meet the set requirements 1.每班用OK，NG样品校正测漏设备。 2.设备自动监控测量气压，不能满足设定要求时报警	1. PQC patrol inspection according to the process checking instruction 2. The automatic monitoring function of the equipment is checked before each production 1.PQC按照过程检查指引做巡视检查 2.设备自动监控功能每次生产前点检	4	84	None 无						

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS (潜在失效模式及影响分析)

Part Description(产品名称) : 后端盖

Part No产品编号:F01RXXXXXX-XXXX

FMEA's Issue No.编号: FM-XXX-1

Model /Vehicle 车型/车辆 :

Key Date关键日期: 2022-12-23

Date修改日期: 2023/1/10 Rev版本: 01

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Core Functional Team多功能小组: *** *** *** *** *** *** *** *** *** *** *** *** *** *** *** ***

Prepared By 编制者: ***

CC副本:

ID	Process Step / Function 过程步骤/功能	Requirement 要求	Potential Failure Mode 潜在失效模式	Potential Effect(s) of Failure 潜在失效影响	Severity 严重度	Classification 分类	Potential Cause(s) of Failure 潜在失效原因	Occurrence 频度	Current Process Controls Prevention 现行过程控制 预防	Current Process Controls Detection 现行过程控制 探测	Detection 探测度	RPN 风险顺序数	Recommended Action 推荐措施	Responsibility & Target Completion Date 职责和目标完成日期	Action Results实施结果				
															Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
		Leakage test machine can identify the part staute exactly 测漏设备能准确识别产品	The defactive goods were judged to be qualified 不良品识别为良品	Effect customer usage. 影响客户使用功能	7		1. The leak betector misjudged. 1. 测漏设备误判 2. Gas contains water 2. 测量气体有水气 3. The air passage is blocked. 3. 通气孔堵塞	3	1.proofread leakage machine by go and no-go part ,record the leakage volume per shift. 1.每班用OK，NG样品校正测漏设备。 2.Compressde air drying treatment . 2.压缩空气干燥处理。 3.Rgular spot inspection and maintenance 3.定期点检维护	with strict stabdard. 加严标准	4	84	None 无						
		Leakage test machine can identify the part staute exactly 测漏设备能准确识别产品	The qualified goods were judged to be defactive 良品识别为不良品	Part scrap in supplier 零件在内部报废	5		1. The leak betector misjudged. 1. 测漏设备误判 2. seal ring damage. 2. 密封圈破损	3	1.proofread leakage machine by go and no-go part ,record the leakage volume per shift . 1.每班用OK，NG样品校正测漏设备。 2.Rgular spot inspection and maintenance 2.定期点检维护2.定期点检维护.	N\A	4	60	None 无						
		after leakage test the part didn't have any deformation 测漏后的产品没有任何的变形	after leakage test the part have deformation 产品变形	Affect function and assembly. 影响功能和装配。	6		1. Fixture limit failure 1. 夹具限位失效	4	1.Rgular spot inspection and maintenance 1.定期点检维护2.定期点检维护.	Operator 100% visual inspection 操作员100%检查	4	96	None 无						
		after leakage test the part surface is within the drawing requirement. 测漏后产品的表面外观符合图纸要求	Scratch on the surface is out of drawing requirement 表面划伤超出图纸要求	Effect the appearance 影响外观	6		Impact between parts when place 产品相互碰撞	3	Accord to visual instruction 按外观检查标准	Operator 100% visual inspection 操作员100%检查	4	72	None 无						
120	Assembling 组装	The pin is not leaky 销钉不漏装	The pin is missing 销钉漏装	内部返工	5		Press-mounting sensor failed 压装感应器失效	2	1. Daily calibration of pressure inductor 2，100% automatic inspection after assembly 1.压装感应器日常校准 2.组装完成后设备100%自动检测	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查 3.100%包装前确认	5	50	None 无						
		Pin in place 销钉装到位	Pressed to the incorrect position 安装不到位	Impact on product function 影响产品功能	7		Press-mounting sensor failed 压装感应器失效	2	1. Daily calibration of pressure inductor 2，100% automatic inspection after assembly 1.压装感应器日常校准 2.组装完成后设备100%自动检测	Regularly calibrate the self-test induction of machine operation 每班记录感应情况，定期校准感应器	5	70	None 无						
130	primer 底涂	Don't miss the glue 不漏底涂	Missing glue 漏底涂	Impact on product function 影响产品功能	7		Leaky operation漏作业	2	Apply dot to designated area after base coating 底涂完成后在指定部位打点	1. 100% visual inspection 1. 100%目视检查 2. Patrol check 2. PQC抽查	3	42	None 无						
140	Appearance Inspection 外观全检	1. Parts don't have dirty 产品没有脏污	1. Parts have dirty 产品脏污	1.Can't meet cleanliness appearance requirement, affect the assembly and performance 1.达不到清洁度要求,影响客户装配和性能	4		The gloves are dirty and not replaced in time Work without gloves Employees are not familiar with standards 手套脏污未及时更换 未戴手套作业 员工不熟悉标准	3	1.Wear gloves before operating and keep golves clean 1.操作前须戴手套并保证手套清洁 2.Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工 3.Place direct label in proper way 不良品位置放置标识	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	60	None 无						
		No knife mark on the machine surface 加工面无刀纹	knife mark on the machined surface imcompletely 加工面有明显刀纹	Affect the product performance 影响产品的使用性能	5		Operators are not familiar with the standard. 员工不熟悉标准	3	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	75	None 无						
		No scratch on the machining surface 加工表面无划痕	Scratch on the machining surface 加工表面划伤	Affect the product performance 影响产品的性能	6		Operators are not familiar with the standard. 员工不熟悉标准	3	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	90	None 无						

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS (潜在失效模式及影响分析)

Part Description(产品名称) : 后端盖 | Part No产品编号:F01RXXXXXX-XXXX | FMEA's Issue No.编号: FM-XXX-1

Model /Vehicle 车型/车辆 : | Key Date关键日期: 2022-12-23 | Date修改日期: 2023/1/10 | Rev版本: 01

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Core Functional Team多功能小组: *** | Prepared By 编制者: ***

CC副本:

ID	Process Step / Function 过程步骤/功能	Requirement 要求	Potential Failure Mode 潜在失效模式	Potential Effect(s) of Failure 潜在失效影响	Severity 严重度	Classification 分类	Potential Cause(s) of Failure 潜在失效原因	Occurrence 频度	Current Process Controls Prevention 现行过程控制 预防	Current Process Controls Detection 现行过程控制 探测	Detection 探测度	RPN 风险顺序数	Recommended Action 推荐措施	Responsibility & Target Completion Date 职责和目标完成日期	Action Results实施结果				
															Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
		No burrs on the function surface 产品面无毛刺	Burrs on function surface. 产品面有毛刺。	Affect the product performance 影响产品的性能	6		Operators are not familiar with the standard. 员工不熟悉标准	3	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	90	None 无						
		气孔满足出货规程要求	密封面气孔超出0.7mm要求，非密封加工面超出1mm	Affect the product performance 影响产品的性能	7		Operators are not familiar with the standard. 员工不熟悉标准	3	Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	105	None 无						
		Parts don't have the rust 产品没有发霉	1. Parts have the rust 产品发霉	Affect the product performance 影响产品的性能	6		The product is not thoroughly dried Employees are not familiar with standards 产品未彻底烤干 员工不熟悉标准	3	1.Wear gloves before operating and keep golves clean 1.操作前须戴手套并保证手套清洁 2.Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	90	None 无						
		There is no yellowish mark on the product 产品没有发黄	There is no yellowish mark on the product 产品发黄	Out of customer's standard it'll be rejected 不符合客户外观标准，客户拒收	5		Operators are not familiar with the standard. 员工不熟悉标准	3	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	75	None 无						
		There is no problem with assembly. 装配没有问题	Missing pin, missing glue 漏装销钉，漏底涂	affect the assembly and product performance 影响产品组装和性能	6		Operators are not familiar with the standard. 员工不熟悉标准	3	Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	90	None 无						
		The bushing is coated with anti-rust oil 衬套涂防锈油	The bushing is missing anti-rust oil 衬套漏涂防锈油	Bushing leakage anti-rust oil bushing rust, customer rejected 衬套生锈，客户拒收	6		Operator omission 作业员遗漏	3	Train operators to apply anti-rust oil according to the designated area 培训作业员，按指定区域进行涂防锈油	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	90	None 无						
150	Packaging 包装	correct quantity 数量正确	lack of product 少数	Customer complains 客户投诉	5		Not full carton 未装满箱	2	1.Training the operator of packing instruction for XXX-XXXX 1.按XXX-XXXX指引培训操作员 2.扫码包装，设备自动检测数量并确认产品的打码及相关信息	100% visual inspection 100% 目视检查 设备自动确认数量100%	5	50	None 无						
		Stick the right label on Package 包装上标签正确	The wrong label on Package 包装上标签错误	Customer complains 客户无法签收，客户投诉	5		Operator is not familiar with the part 操作工错放其他产品标签	2	1.Training the operator of packing instruction for XXX-XXXX 1.按XXX-XXXX指引培训操作员	100% visual inspection by operator&spot check by OQC. 操作工100%目视检查，OQC确认	5	50	None 无						
		Use the card board correctly 正确使用卡板	Use the wrong card board 使用错误卡板	The cartons are damaged and the products are crushed 纸箱破损，产品压坏	5		The box protruded the card board, and the carton was crushed when the board was stacked, and the product was crushed 箱子突出卡板，叠板时压坏纸箱，产品压坏	2	1.Training the operator of packing instruction for XXX-XXXX 1.按XXX-XXXX指引培训操作员	100% visual inspection by operator&spot check by OQC. 操作工100%目视检查，OQC确认	5	50	None 无						
Q2	OQC Inspection OQC 检查	1. Parts don't have dirty 产品没有脏污	1. Parts have dirty 产品脏污	1.Can't meet cleanliness appearance requirement, affect the assembly and performance 1.达不到清洁度要求,影响客户装配和性能	4		The gloves are dirty and not replaced in time Work without gloves Employees are not familiar with standards 手套脏污未及时更换 未戴手套作业 员工不熟悉标准	3	1.Wear gloves before operating and keep golves clean 1.操作前须戴手套并保证手套清洁 2.Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工 3.Place direct label in proper way 不良品位置放置标识	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	60	None 无						
		No knife mark on the machine surface 加工面无刀纹	knife mark on the machined surface imcompletely 加工面有明显刀纹	Affect the product performance 影响产品的使用性能	5		Operators are not familiar with the standard. 员工不熟悉标准	2	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	50	None 无						

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS (潜在失效模式及影响分析)

Part Description(产品名称) : 后端盖

Part No产品编号:F01RXXXXXX-XXXX

FMEA's Issue No.编号: FM-XXX-1

Model /Vehicle 车型/车辆 :

Key Date关键日期: 2022-12-23

Date修改日期: 2023/1/10 Rev版本: 01

Designer设计責任人:

D/PFMEA Date设计/过程制作日期: 2022-8-19

Page 业数: 11

Core Functional Team多功能小组: *** *** *** *** *** *** *** *** *** *** *** *** *** *** *** ***

Prepared By 编制者: ***

CC副本:

ID	Process Step / Function 过程步骤/功能	Requirement 要求	Potential Failure Mode 潜在失效模式	Potential Effect(s) of Failure 潜在失效影响	Severity 严重度	Classification 分类	Potential Cause(s) of Failure 潜在失效原因	Occurrence 频度	Current Process Controls Prevention 现行过程控制 预防	Current Process Controls Detection 现行过程控制 探测	Detection 探测度	RPN 风险顺序数	Recommended Action 推荐措施	Responsibility & Target Completion Date 职责和目标完成日期	Action Results实施结果				
															Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
		No scratch on the machining surface 加工表面无划痕	Scratch on the machining surface 加工表面划伤	Affect the product performance 影响产品的性能	6		Operators are not familiar with the standard. 员工不熟悉标准	2	1.签订限度样品 2.Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	60	None 无						

POTENTIAL FAILURE MODE AND EFFECTS ANALYSIS (潜在失效模式及影响分析)

Part Description(产品名称) : 后端盖

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															Actions Taken & Effective Date 采取的措施和有效日期	Severity 严重度	Occurrence 频度	Detection 探测度	RPN 风险顺序数
		No burrs on the function surface 产品面无毛刺	Burrs on function surface. 产品面有毛刺。	Affect the product performance 影响产品的性能	6		Operators are not familiar with the standard. 员工不熟悉标准	2	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	60	None 无						
		气孔满足出货规程要求	密封面气孔超出0.7mm要求，非密封加工面超出1mm	Affect the product performance 影响产品的性能	7		Operators are not familiar with the standard. 员工不熟悉标准	2	Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	70	None 无						
		Parts don't have the rust 产品没有发霉	1. Parts have the rust 产品发霉	Affect the product performance 影响产品的性能	6		The product is not thoroughly dried Employees are not familiar with standards 产品未彻底烤干 员工不熟悉标准	2	1.Wear gloves before operating and keep golves clean 1.操作前须戴手套并保证手套清洁 2.Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	60	None 无						
		产品没有发黄	产品发黄	Out of customer's standard it'll be rejected 不符合客户外观标准，客户拒收	5		Operators are not familiar with the standard. 员工不熟悉标准	3	1. 签订限度样品 2. Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	75	None 无						
		There is no problem with assembly. 装配没有问题	Missing pin, missing glue 漏装销钉，漏底涂	affect the assembly and product performance 影响产品组装和性能	6		Operators are not familiar with the standard. 员工不熟悉标准	3	Identify all defects in WI and train the operator. 制作作业指导书中标明所有缺陷并培训员工	1.100% visual inspection 1.100%目视检查 2. Patrol check 2.PQC抽查	5	90	None 无						
160/170	Storage and Delivery 入库并交付	No damage on packing 包装没损坏	Packing damage 包装损坏	the product damage, affect the assembly 产品碰伤影响装配	6		The box fell to the ground and the board turned upside down 箱子掉地，倒板	2	1.Training the operator of packing instruction for XXX-XXXX 1.按XXX-XXXX指引培训操作员	100% visual inspection by operator&spot check by OQC. 操作工100%目视检查，OQC确认	5	60	None 无						
		Stick the right label on Package 包装上标签正确	The wrong label on Package 包装上标签错误	Customer complains 客户无法签收，客户投诉	5		Operator is not familiar with the part 操作工错放其他产品标签	2	1.Training the operator of packing instruction for XXX-XXXX 1.按XXX-XXXX指引培训操作员	100% visual inspection by operator&spot check by OQC. 操作工100%目视检查，OQC确认	5	50	None 无						
		correct quantity 数量正确	lack of product 少数	Customer complains 客户投诉	5		Packing pallet in wrong place 卡板放置错误	2	Placing pallet in given place 分区域放置产品	100% visual inspection 100% 目视检查 设备自动确认数量100%	5	50	None 无						
		correct loading 正确装车	incorrct loading 不正确装车	Customer complains 客户投诉	5		reloading 倒板	2	1.Training the operator of packing instruction for XXX-XXXX 1.按XXX-XXXX指引培训操作员	100% visual inspection by operator&spot check by OQC. 操作工100%目视检查，OQC确认	5	50	None 无						

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